

CALC BOY

User Manual



Version 3.1.0 - Generated from the current repository state.

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Introduction

CALC BOY is a local-first PWA calculator styled after classic handheld consoles. This documentation describes the current source code for version 3.1.0.

Tip: The display is also interactive. Tap it for history; long-press it to copy the current result.

Installation

- Open the live link in Safari or Chrome.
- Use Share or the browser menu and choose Add to Home Screen.
- Launch CALC BOY from the home screen. Installed mode opens the app fullscreen.

Offline Mode

The service worker caches the app shell, index.html and icons under calcboy-v3.1.0. Registration runs only on HTTPS. After the first online launch, the app can start from cache.

First calculation



Example: $12 + 30 =$ shows 42 and adds the expression to history.

History, copy and share



The last ten successful calculations are saved. The history view also shows sum and average. Text and PNG export are available when the browser supports sharing or clipboard APIs.

BASIC



Examples

Example	Steps
Smart percent	$100 + 10 \% = 110$

Buttons

Key	Function	Key	Function
AC	Clears entry and active operation; in RPN it also clears the stack.	*	Runs the operation shown on the key.
+/-	Runs the operation shown on the key.	1	Runs the operation shown on the key.
%	Smart percent: with + or - it means percent of the first operand.	2	Runs the operation shown on the key.
sqrt	Applies the scientific function to the displayed value.	3	Runs the operation shown on the key.
7	Runs the operation shown on the key.	-	Runs the operation shown on the key.
8	Runs the operation shown on the key.	0	Runs the operation shown on the key.
9	Runs the operation shown on the key.	,	Runs the operation shown on the key.
/	Runs the operation shown on the key.	+	Runs the operation shown on the key.
4	Runs the operation shown on the key.	EXT	Navigates to this calculator page.
5	Runs the operation shown on the key.	=	Runs the operation shown on the key.
6	Runs the operation shown on the key.		

EXT



Examples

Example	Steps
DEG/RAD	DEG: 30 sin = 0.5
Memory	42 M+ stores 42; MR recalls it.

Buttons

Key	Function	Key	Function
MC	Runs the operation shown on the key.	x^2	Applies the scientific function to the displayed value.
MR	Runs the operation shown on the key.	x^3	Applies the scientific function to the displayed value.
M+	Runs the operation shown on the key.	x^y	Applies the scientific function to the displayed value.
M-	Runs the operation shown on the key.	cbrt	Runs the operation shown on the key.
sin	Applies the scientific function to the displayed value.	1/x	Runs the operation shown on the key.
cos	Applies the scientific function to the displayed value.	n!	Applies the scientific function to the displayed value.
tan	Applies the scientific function to the displayed value.	pi	Applies the scientific function to the displayed value.
DEG/RAD	Toggles saved angle mode for trigonometry.	e	Applies the scientific function to the displayed value.
log	Applies the scientific function to the displayed value.	BASIC	Navigates to this calculator page.
ln	Applies the scientific function to the displayed value.	MEHR	Cycles EXT -> CONV -> FIN -> PRG -> PLOT -> FORM.
10^x	Applies the scientific function to the displayed value.	=	Runs the operation shown on the key.
e^x	Applies the scientific function to the displayed value.		

CONV



Examples

Example	Steps
VAT	100 MW+19 = 119; 119 MW-19 = 100

Buttons

Key	Function	Key	Function
km->mi	Runs the operation shown on the key.	km/h->mph	Runs the operation shown on the key.
mi->km	Runs the operation shown on the key.	mph->km/h	Runs the operation shown on the key.
m->ft	Runs the operation shown on the key.	h->min	Runs the operation shown on the key.
ft->m	Runs the operation shown on the key.	min->h	Runs the operation shown on the key.
C->F	Runs the operation shown on the key.	MW+19	Runs the operation shown on the key.
F->C	Runs the operation shown on the key.	MW-19	Runs the operation shown on the key.
kg->lb	Runs the operation shown on the key.	MW+7	Runs the operation shown on the key.
lb->kg	Runs the operation shown on the key.	MW-7	Runs the operation shown on the key.
cm->in	Runs the operation shown on the key.	BASIC	Navigates to this calculator page.
in->cm	Runs the operation shown on the key.	MEHR	Cycles EXT -> CONV -> FIN -> PRG -> PLOT -> FORM.
l->gal	Runs the operation shown on the key.	=	Runs the operation shown on the key.
gal->l	Runs the operation shown on the key.		

FIN



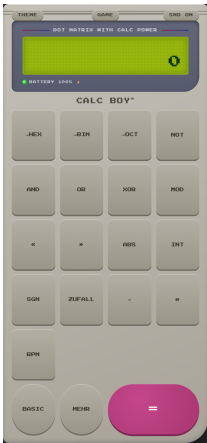
Examples

Example	Steps
Compound interest	K0 1000, P 3, years 10, rate 50 -> ENDWERT.
Bill split	Set persons to 4, enter 80, press / PERS -> 20.

Buttons

Key	Function	Key	Function
SET K0	Stores the displayed value for this parameter.	TIP+20%	Runs the operation shown on the key.
SET P%	Stores the displayed value for this parameter.	SET PERS	Stores the displayed value for this parameter.
SET JAHRE	Stores the displayed value for this parameter.	/ PERS	Runs the operation shown on the key.
SET RATE/M	Stores the displayed value for this parameter.	SET KURS	Stores the displayed value for this parameter.
INFO	Shows stored parameters or variables.	EUR->\$	Runs the operation shown on the key.
ENDWERT	Runs the operation shown on the key.	\$->EUR	Runs the operation shown on the key.
ZINSEN	Runs the operation shown on the key.	BASIC	Navigates to this calculator page.
RESET	Restores finance defaults.	MEHR	Cycles EXT -> CONV -> FIN -> PRG -> PLOT -> FORM.
TIP+10%	Runs the operation shown on the key.	=	Runs the operation shown on the key.
TIP+15%	Runs the operation shown on the key.		

PRG



Examples

Example	Steps
Binary operations	5 AND 3 = 1; 5 OR 2 = 7.
RPN	Enter 12 =, enter 3, press + -> 15.

Buttons

Key	Function	Key	Function
->HEX	Runs the operation shown on the key.	ABS	Runs the operation shown on the key.
->BIN	Runs the operation shown on the key.	INT	Runs the operation shown on the key.
->OCT	Runs the operation shown on the key.	SGN	Runs the operation shown on the key.
NOT	Runs the operation shown on the key.	ZUFALL	Runs the operation shown on the key.
AND	Runs the operation shown on the key.	pi	Applies the scientific function to the displayed value.
OR	Runs the operation shown on the key.	e	Applies the scientific function to the displayed value.
XOR	Runs the operation shown on the key.	RPN	Toggles Reverse Polish Notation mode.
MOD	Runs the operation shown on the key.	BASIC	Navigates to this calculator page.
<<	Runs the operation shown on the key.	MEHR	Cycles EXT -> CONV -> FIN -> PRG -> PLOT -> FORM.
>>	Runs the operation shown on the key.	=	Runs the operation shown on the key.

PLOT



Examples

Example	Steps
Graph	Press sin x to draw $y = \sin x$.

Buttons

Key	Function	Key	Function
sin x	Applies the scientific function to the displayed value.	sinh	Applies the scientific function to the displayed value.
cos x	Applies the scientific function to the displayed value.	cosh	Applies the scientific function to the displayed value.
tan x	Applies the scientific function to the displayed value.	tanh	Applies the scientific function to the displayed value.
x^2	Applies the scientific function to the displayed value.	x^4	Applies the scientific function to the displayed value.
x^3	Applies the scientific function to the displayed value.	GAUSS	Runs the operation shown on the key.
sqrt x	Applies the scientific function to the displayed value.	FLOOR	Runs the operation shown on the key.
log x	Applies the scientific function to the displayed value.	sinc x	Applies the scientific function to the displayed value.
1/x	Runs the operation shown on the key.	$x \cdot \sin x$	Applies the scientific function to the displayed value.
e^x	Applies the scientific function to the displayed value.	BASIC	Navigates to this calculator page.
ln x	Applies the scientific function to the displayed value.	MEHR	Cycles EXT -> CONV -> FIN -> PRG -> PLOT -> FORM.
x	Runs the operation shown on the key.	=	Runs the operation shown on the key.
2^x	Applies the scientific function to the displayed value.		

FORM



Examples

Example	Steps
Formula assistant	SET A=20, SET B=150, % VON -> 30.

Buttons

Key	Function	Key	Function
SET A	Stores the displayed value for this parameter.	BMI	Runs the operation shown on the key.
SET B	Stores the displayed value for this parameter.	AVG ABC	Runs the operation shown on the key.
SET C	Stores the displayed value for this parameter.	CLR ABC	Runs the operation shown on the key.
INFO	Shows stored parameters or variables.	KM/H	Runs the operation shown on the key.
% VON	Runs the operation shown on the key.	NET19	Runs the operation shown on the key.
DREISATZ	Runs the operation shown on the key.	BRU19	Runs the operation shown on the key.
KREIS A	Runs the operation shown on the key.	BASIC	Navigates to this calculator page.
KREIS U	Runs the operation shown on the key.	MEHR	Cycles EXT -> CONV -> FIN -> PRG -> PLOT -> FORM.
PYTH	Runs the operation shown on the key.	=	Runs the operation shown on the key.
OHM	Runs the operation shown on the key.		

Plotting



A plot replaces the LCD with a pixel graph. Press any non-plot key or tap the display to close it.

Games



Math Attack asks arithmetic questions for 30 seconds. Snake uses the number pad 2/4/6/8 or arrow keys and stores its own high score.

Landscape mode



In landscape, the scientific shortcut column is visible beside BASIC, giving quick access to sin, cos, tan, log, ln, powers, pi and e.

Keyboard

Key	Action
0-9	digits
, or .	decimal separator
+ - * /	basic operators
Enter or =	equals
Escape	AC
%	percent
r or w	square root
arrow keys	Snake direction and secret-code input

Storage and Privacy

All data remains in browser localStorage. There is no analytics code, no external font request and no live exchange-rate API.

theme, sound, angle mode, history, memory, finance parameters, exchange rate, persons, RPN mode, RPN stack, formula variables, game high scores, Virtual Boy unlock.

Limitations

Service worker registration only runs on HTTPS. The inline manifest has icons only when opened via HTTP or HTTPS. Currency conversion uses a manually stored rate and no live-rate API. Programmer base conversion displays the converted value in the status line; it does not replace the main display. Browser sharing, clipboard, vibration and battery display depend on browser support. The UI language inside the app is German.

Troubleshooting

If an old version appears after update, close and reopen the app twice. If sound does not play, tap once inside the app and check SND ON. If sharing is unavailable, use clipboard copy or browser download fallback. To reset all data, delete website data for this domain.

Version Information

CALC BOY 3.1.0 - calcboy-v3.1.0 - GPL-3.0-or-later